MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: Mansourah Massarah Gold Project

Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION

3100 CRUSHING

DESIGN CALCULATION OF BAG FILTER & DUCT SUPPORT FOUNDATION

AS BUILT

MD-216-3100-EG-CV-CAL-2023

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
D01	12-10-2020	AS BUILT	APL	SBY	PRI	
C01	21-10-2020	ISSUED FOR CONSTRUCTION	APL	SBY	PRI	
A01	18-08-2020	FIRST ISSUE	APL	SBY	PRI	

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Ma'aden Document Number:							
MD-216-3100-EG-CV-CAL-2023							D01
Outotec Document Number:							
900707	GAU01	ZC01	CED01	31004	OU100692094		2
L&T Document Number:							
O19045	080	C	K	24366	004		2

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Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	18-08-2020	21-10-2020	12-10-2023				Date	18-08-2020	21-10-2020	12-10-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUC	AS BUILT		
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Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2007	Saudi Code for Concrete Structures

TABLE OF CONTENTS

DEFINITIONS AND ABBREVIATIONS 3

1 INTENT 4

2 EXECUTIVE SUMMARY 4

3 SCOPE 4

4 CONCRETE STRUCTURE 4

4.1 Design 4
4.2 Materials 4
4.3 Reference 4

5 INPUT LOADS FROM SUPERSTRUCTURE 4

5.1 Support Reaction from superstructure 4
5.2 Load Combinations 5

6 GEOMETRY 5

6.1 Geometry of Foundation arrangement 5

7 DESIGN OF CONCRETE COMPONENTS 6-12

7.1 Design summary of Pedestals 6
7.2 Design summary of Tie Beams 6
7.3 Design summary of Foundations 6

ATTACHMENT 1 MECHANICAL GENERAL ARRANGEMENT DRAWINGS 13

ATTACHMENT 2 STRUCTURAL GENERAL ARRANGEMENT DRAWINGS CONTAINING BASE PLATE DETAILS 22

ATTACHMENT 3 RELATED CIVIL GA & RC DRAWINGS OF NEARBY STRUCTURE 27

ATTACHMENT 4 STAAD INPUT OF TOTAL FOUNDATION ARRANGEMENT 33

ATTACHMENT 5 STAAD OUTPUT OF TOTAL FOUNDATION ARRANGEMENT 46

DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
RL	:	Reduced Level
C.L.	:	Centre Line
CJ	:	Construction Joint
FDN	:	Foundation
TOC	:	Top of concrete
FGL	:	Finished Ground Level
FFL	:	Finished Floor Level
BOF	:	Bottom of footing
TYP	:	Typical
TOC	:	Top of concrete
IP	:	Insert Plate
RCC	:	Reinforced Cement Concrete
MKD.	:	Marked
THK.	:	Thick

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1. INTENT

The document is prepared for foundation design calculation of Deducting duct support, Bag filter and its aceess staircase.
This bag filter is between Secondary Crusher House and Tertiary Crusher House.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.
MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.
The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

This document is prepared for foundation design of Deducting duct support, Bag filter and its aceess staircase. All the loads applied to the structure for analysis & design followed Civil -Structural Design criteria and SBC 301:2007. The column and beam reinforcement design have been performed in Staad-Pro Software. The design of combined footings have been performed in MS Excel.
Total foundation scheme has been modelled and analysed using FEM analysis in Staad-Pro Software.

4. CONCRETE STRUCTURE

4.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304 and ACI 318M.

4.2 Materials

Concrete Strength of 28 days
· Foundation and Structural Members: 35 MPa
· Grade slab and pavement: 30 MPa
· Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade
· Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
· Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

4.3 Foundations for Structures

Foundation shall be Isolated and combined Shallow Concrete foundation. The design calculations are prepared as per recommendation of standard SBC-304 Clause 13.3.

4.4 Minimum Concrete Cover

Items	Minimum Cover(mm)
Foundation (Side & Bottom)	75
Foundation (Top)	50
Column & Pedestal	50
Beam	50
Slab	40

4.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 500 kN/m2 at 3.5m depth below FGL (ie. RL 909.000m) refer Table 4 - BH-17,18,19,25,29 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.6 Reference

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

5 INPUT LOADS FROM SUPERSTRUCTURE

5.1 Support Reaction from superstructure

For load calculation and support reaction of superstructure, refer Document No: MD-216-3120-EG-ST-CAL-2021
For support reaction of Bag filter superstructure, refer Drawing No: MD-216-3100-EG-ME-DED-V509-2001

	Bag Filter Structure		
	Due to Wind (in KN)	Due to Seismeic (in KN)	Guiding LC
Total load along +X direction	566	270	Wind
Total load along -X direction	567	270	Wind
Total load along +Z direction	615	270	Wind
Total load along -Z direction	615	270	Wind

As total base shear due to seismic force in each direction is much lesser than the wind load in respective direction. So, in foundation design load combination considering seismic loads are not considered.

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5.2 LOAD COMBINATIONS

5.2.1 Load Combination for Base pressure calculation

(As per Section 2.4 of SBC 301 -2007)

Load Cases	Reference
LOAD COMB 201 1.0 (DL + LL)	(Eq. 2.4.1-1 of SBC 301)
LOAD COMB 202 1.0 DL+ 0.75 WL(+X) + 0.75 LL LOAD COMB 203 1.0 DL+ 0.75 WL(-X) + 0.75 LL LOAD COMB 204 1.0 DL+ 0.75 WL(+Z) + 0.75 LL LOAD COMB 205 1.0 DL+ 0.75 WL(-Z) + 0.75 LL	(Eq. 2.4.1-6 of SBC 301)
LOAD COMB 206 0.6 DL + 1.0 WL(+X) LOAD COMB 207 0.6 DL + 1.0 WL(-X) LOAD COMB 208 0.6 DL + 1.0 WL(+Z) LOAD COMB 209 0.6 DL + 1.0 WL(-Z)	(Eq. 2.4.1-7 of SBC 301)
LOAD COMB 210 1.0 (DL + TL1) + 1.0 LL LOAD COMB 211 1.0 (DL + TL2) + 1.0 LL	(Eq. 2.4.1-2 of SBC 301)

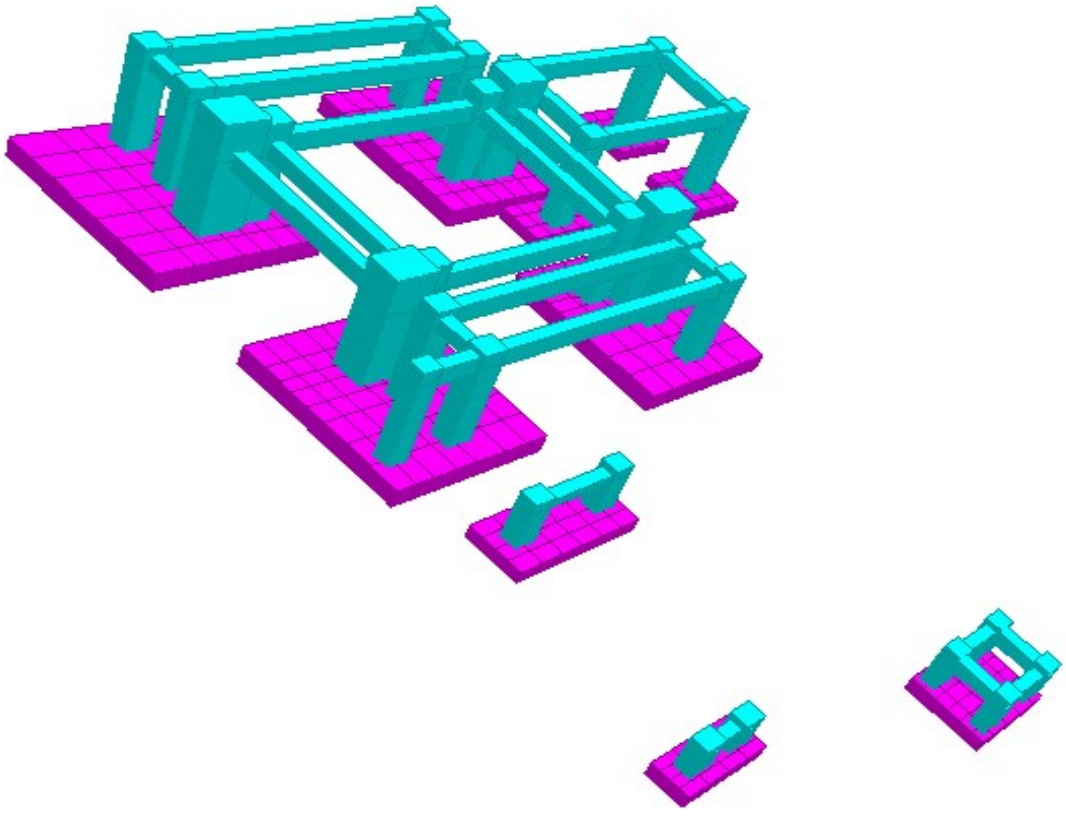
5.2.2 Load Combination for Concrete design

Load Case	Reference
LOAD COMB 301 1.4 DL + 1.7 LL	(Eq. 2.3.2-1 of SBC 301)
LOAD COMB 302 1.2 DL+ 1.6 WL(+X) + 0.5 LL LOAD COMB 303 1.2 DL+ 1.6 WL(-X) + 0.5 LL LOAD COMB 304 1.2 DL+ 1.6 WL(+Z) + 0.5 LL LOAD COMB 305 1.2 DL+ 1.6 WL(-Z) + 0.5 LL	(Eq. 2.3.2-4 of SBC 301)
LOAD COMB 306 0.9 DL + 1.6 WL(+X) LOAD COMB 307 0.9 DL + 1.6 WL(-X) LOAD COMB 308 0.9 DL + 1.6 WL(+Z) LOAD COMB 309 0.9 DL + 1.6 WL(-Z)	(Eq. 2.3.2-6 of SBC 301)
LOAD COMB 310 1.4 (DL + TL1) + 1.7 LL LOAD COMB 311 1.4 (DL + TL2) + 1.7 LL	(Eq. 2.3.2-2 of SBC 301)

Eq. 2.3.2-3 of SBC 301 - 2007 has not been considered as it would not produce worst load combination in the absence of Roof Live Load & Rain Load

6 GEOMETRY

6.1 Geometry of Foundation arrangement



Isometric view of total foundation arrangement (From Staad)

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7. DESIGN OF CONCRETE COMPONENTS

All foundations have been designed using worksheet in accordance to SBC-304.

All beams & columns have been designed in Staad-Pro by ACI 318M.

For Design Results, Refer Attachemnt 5 - Staad Output file.

7.1 Design Summary of Column

7.1.1 Column Reinforcement Summary

Column mark	Width	Depth	Reinforcement	Stirrup	Remarks	Numbers
PD1	450	450	8 No.- D20	3 Nos D10 @ 250c/c	Pedestals for duct support	9
PD2	500	700	12 No.- D20	3 Nos D10 @ 250c/c	Pedestals for duct support	8
PC1	600	600	12 No.- D25	3 Nos D10 @ 250c/c	Pedestals for conv support	4
PB1	900	1000	24 No.- D25	6 Nos D10 @ 250c/c	Pedestals for Bag filter support	4
PS1	550	550	12 No.- D20	3 Nos D10 @ 250c/c	Pedestals for duct support	4

7.2 Design Summary of Beam

7.2.1 Beam Reinforcement Summary

Beam mark	Width (mm)	Depth (mm)	Top Reinforcement	Bottom Reinforcement	Stirrup	Numbers
TB1	300	300	3 Nos D12	3 Nos D12	2L D10 STRPS @ 175c/c	4
TB2	300	400	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	6
TB3	300	600	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	6
TB4	300	600	3 Nos D20	3 Nos D20	2L D10 STRPS @ 250c/c	2
TB5	300	500	3 Nos D20	3 Nos D20	2L D10 STRPS @ 250c/c	4
TB6	275	500	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	2

7.3 Design of Foundation

7.3.1 Foundation marking, Max. Base pressure and Reinforcement Summary

Structure Name	Foundation mark	Length (Along vertical) (mm)	Width (Along horizontal) (mm)	Thickness (mm)	Max pressure (kN/m^2)	Top reinforcement	Bottom reinforcement	F.O.S.	Numbers
Bag filter stair support	F1	1800	1800	400	127	D12 @ 200c/c B/W	D16 @ 200c/c B/W	**0.25**	3
TR1 OF Duct support	CF1	2200	2200	400	49	D12 @ 200c/c B/W	D16 @ 200c/c B/W	**0.09**	1
TR2 OF Duct support	CF2	1000	1950	400	49	D12 @ 200c/c B/W	D16 @ 200c/c B/W	**0.09**	1
TR3 OF Duct support	CF3	1500	3000	400	49	D12 @ 200c/c B/W	D16 @ 200c/c B/W	**0.09**	1
TR4 OF Duct supp. & Bag filter supp.	CF4	5000	3800	600	190	D16 @ 150c/c B/W	D20 @ 150c/c B/W	**0.34**	1
TR4 OF Duct supp. & Bag filter supp.	CF4A	5000	3800	600	190	D16 @ 150c/c B/W	D20 @ 150c/c B/W	**0.34**	1
TR5 OF Duct supp. & Bag filter supp.	CF5	5000	3800	600	190	D16 @ 150c/c B/W	D20 @ 150c/c B/W	**0.34**	1
TR5 OF Duct supp. & Bag filter supp.	CF5A	5000	3800	600	190	D16 @ 150c/c B/W	D20 @ 150c/c B/W	**0.34**	1

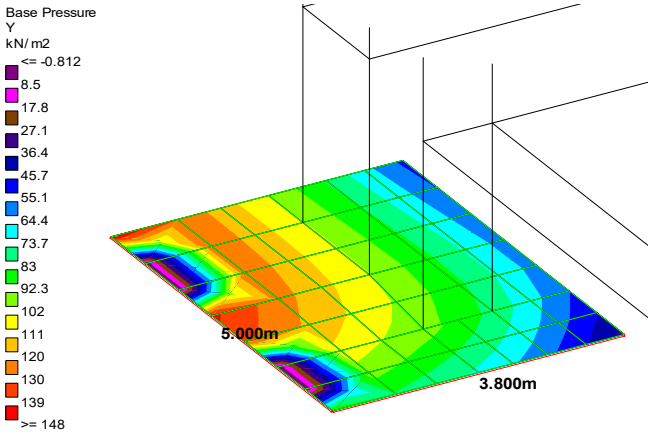
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7.3.2 Design of combined footing mkd. CF4, CF4A, CF5 & CF5A

For Calculation of Base pressure and design, Refer STAAD File: 3210-CNV-001_TR1 FDN MKD_Combined.std

Staad Supoort Nodes are 5 & 6



Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +912.750m)
Finished Ground level	F.G.L	=	0.0	m	(Corresponds to RL +912.250m)
Bottom of foundation	B.O.F	=	-3.50	m	(Corresponds to RL +910.000m)
Top of pedestal (staad support)	T.O.F	=	0.35	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	3.800	m	
Width of footing	Lx	=	5.000	m	
Thickness of footing	D	=	0.60	m	
Depth of pedestal	Bx	=	0.60	m	
Width of pedestal	Bz	=	0.60	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	563	KN/m^2	
Soil load on footing in staad		=	52.2	KN/m^2	staad input

1) Check for Gross Base Pressure :

Maximum Base Pressure		=	190.0	KN/m^2	Load Case 205
Minimum Base Pressure		=	-0.4	KN/m^2	Load Case 206

2) Design of foundation Raft :

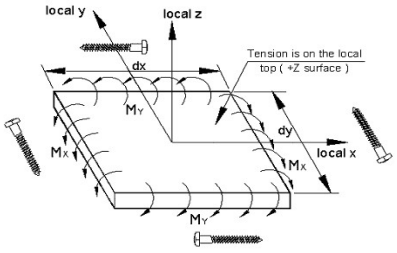
Slab Thickness =	600 mm	Yield Strength of Main Reinforcement , fy =	420 N/mm^2
Width =	1000 mm	Yield Strength of Shear Reinforcement , fyt =	420 N/mm^2
Clear Cover =	75 mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9
Compressive Strength of Concrete , fc' =	35 N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : MX,MY)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	65.905	kN-m	Corresponding Mxy	-24.752	kN-m	*(Plate No. - 666 , L/C No. - 306)*
Min. Mx	-298.655	kN-m	Corresponding Mxy	28.139	kN-m	*(Plate No. - 633 , L/C No. - 302)*
Max. My	4.893	kN-m	Corresponding Mxy	6.541	kN-m	*(Plate No. - 669 , L/C No. - 307)*
Min. My	-100.224	kN-m	Corresponding Mxy	32.591	kN-m	*(Plate No. - 679 , L/C No. - 303)*

FOR TOP FACE REINFORCEMRNT :



Mx1 =	Mx + abs(Mxy) =	65.905 + 24.752 =	90.657 kN-m
My1 =	My + abs(Mxy) =	4.893 + 6.541 =	11.434 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	65.905 + 125.212 =	191.117 kN-m
My2 =	My+abs(Mxy^2/Mx) =	4.893 + 6.955 =	11.848 kN-m

DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 90.657 kN-m
Myd = 11.434 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 90.657 kN-m

Φ*Kn = Mu/bd^2 deff. = 517 mm

Mu/bd^2 = 90.657X10^6/(1000X517^2) = 0.339 N/mm^2 < 6.9 N/mm^2

Pt = 0.0009 Reqd. Ast = 0.0009X1000X517/100 = 4.65 cm^2

Provided 16 mm @ 150 C/C (Provide Ast = 13.4)cm^2

Total Provide Ast = 13.4 cm^2 O.K.

Total Provide Pt = 0.0026

ALONG LOCAL Y - DIRECTION :

Mu = 11.434 kN-m

Φ*Kn = Mu/bd^2 deff. = 501 mm

Mu/bd^2 = 11.434X10^6/(1000X501^2) = 0.043 N/mm^2 < 6.9 N/mm^2

Pt = 0.0001 Reqd. Ast = 0.0001X1000X517/100 = 0.5 cm^2

Provided 16 mm @ 150 C/C (Provide Ast = 13.4)cm^2

Total Provide Ast = 13.4 cm^2 O.K.

Total Provide Pt = 0.0027

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FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -298.655 - 28.139 = -326.794 kN-m
My1 = My - abs(Mxy) = -100.224 - 32.591 = -132.815 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -298.655 - 7.9 = -306.555 kN-m
My2 = My-abs(Mxy^2/Mx) = -100.224 - 3.557 = -96.667 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -326.794 kN-m
Myd = -132.815 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 326.794 kN-m
Φ*Kn = Mu/bd^2 deff. = 515 mm
Mu/bd^2 = 326.794X10^6/(1000X515^2) = 1.232 N/mm^2 < 6.9 N/mm^2
Pt = 0.0033 Reqd. Ast = 0.0033X1000X515/100 = 17 cm^2
Provided 20 mm @ 150 C/C (Provide Ast = 20.94)cm^2

Total Provide Ast = 20.94 cm^2 O.K.
Total Provide Pt = 0.0041

ALONG LOCAL Y - DIRECTION :

Mu = 132.815 kN-m
Φ*Kn = Mu/bd^2 deff. = 495 mm
Mu/bd^2 = 132.815X10^6/(1000X495^2) = 0.542 N/mm^2 < 6.9 N/mm^2
Pt = 0.0014 Reqd. Ast = 0.0014X1000X495/100 = 6.93 cm^2
Provided 20 mm @ 150 C/C (Provide Ast = 20.94)cm^2

Total Provide Ast = 20.94 cm^2 O.K.
Total Provide Pt = 0.0042

SHEAR CHECK

(Staad Group : SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = -0.627 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 657 , L/C No. - 311)*
Vu = 322.91 kN

Provided Pt = 0.0041
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0041X322.905X10^3X515/326.794/10^6)X1000X515/10^3
= 505.751 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0041)X1000X515/10^3
= 523.38 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X515/10^3
= 883.567 kN
HENCE, Vc = 505.751 kN
Φt*Vc = 0.75X505.751 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 379.313 kN > 322.905T , Hence O.K.

ALONG LOCAL Y - DIRECTION :

SQy = -0.596 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 657 , L/C No. - 311)*
Vu = 295.02 kN

Pt provided = 0.0042
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0042X295.02X10^3X495/132.815/10^3)X1000X495/10^3
= 507.414 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0042)X1000X495/10^3
= 503.897 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X495/10^3
= 849.253 kN
HENCE, Vc = 503.897 kN
Φt*Vc = 0.75X503.897 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 377.923 kN > 295.02T , Hence O.K.

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

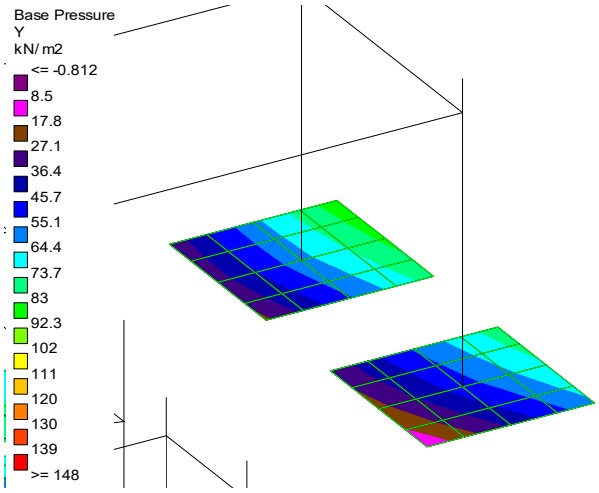
Ma'aden Revision No:	A01	C01	D01
Date	16-04-2020	21-10-2020	12-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	16-04-2020	21-10-2020	12-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

7.3.3 Design of combined footing mkd. F1

For Calculation of Base pressure and design, Refer STAAD File: 3210-CNV-001_TR1 FDN MKD_Combined.std

Staad Supoort Nodes are 5 & 6



Finished Floor level	F.F.L	=	0.3	m	(Corresponds to RL +912.750m)
Finished Ground level	F.G.L	=	0.0	m	(Corresponds to RL +912.250m)
Bottom of foundation	B.O.F	=	-3.50	m	(Corresponds to RL +910.000m)
Top of pedestal (staad support)	T.O.F	=	0.35	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	1.800	m	
Width of footing	Lx	=	1.800	m	
Thickness of footing	D	=	0.40	m	
Depth of pedestal	Bx	=	0.60	m	
Width of pedestal	Bz	=	0.60	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	563	KN/m^2	
Soil load on footing in staad		=	55.8	KN/m^2	staad input

1) Check for Gross Base Pressure :

Maximum Base Pressure		=	127.0	KN/m^2	Load Case 205
Minimum Base Pressure		=	-13.8	KN/m^2	Load Case 209

2) Design of foundation Raft :

Slab Thickness =	400 mm	Yield Strength of Main Reinforcement , fy =	420 N/mm^2
Width =	1000 mm	Yield Strength of Shear Reinforcement , fyt =	420 N/mm^2
Clear Cover =	75 mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9
Compressive Strength of Concrete , fc' =	35 N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75

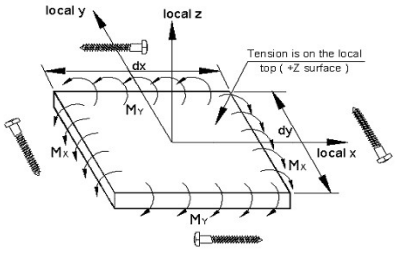
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : MX,MY)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	26.578	kN-m	Corresponding Mxy	-9.938	kN-m	*(Plate No. - 861 , L/C No. - 309)*
Min. Mx	-95.534	kN-m	Corresponding Mxy	12.828	kN-m	*(Plate No. - 861 , L/C No. - 304)*
Max. My	29.847	kN-m	Corresponding Mxy	-7.141	kN-m	*(Plate No. - 864 , L/C No. - 309)*
Min. My	-91.193	kN-m	Corresponding Mxy	-0.834	kN-m	*(Plate No. - 880 , L/C No. - 304)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	26.578 + 9.938 =	36.516 kN-m
My1 =	My + abs(Mxy) =	29.847 + 7.141 =	36.988 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	26.578 + 3.309 =	29.887 kN-m
My2 =	My+abs(Mxy^2/Mx) =	29.847 + 4.631 =	34.478 kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 36.516 kN-m
Myd = 36.988 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 36.516 kN-m

Φ*Kn = Mu/bd^2 deff. = 319 mm

Mu/bd^2 = 36.516X10^6/(1000X319^2) = 0.359 N/mm^2 < 6.9 N/mm^2

Pt = 0.0010 Reqd. Ast = 0.001X1000X319/100 = 3.19 cm^2

Provided 12 mm @ 200 C/C (Provide Ast = 5.65)cm^2

Total Provide Ast = 5.65 cm^2 O.K.

Total Provide Pt = 0.0018

ALONG LOCAL Y - DIRECTION :

Mu = 36.988 kN-m

Φ*Kn = Mu/bd^2 deff. = 307 mm

Mu/bd^2 = 36.988X10^6/(1000X307^2) = 0.363 N/mm^2 < 6.9 N/mm^2

Pt = 0.0010 Reqd. Ast = 0.001X1000X319/100 = 3.07 cm^2

Provided 12 mm @ 200 C/C (Provide Ast = 5.65)cm^2

Total Provide Ast = 5.65 cm^2 O.K.

Total Provide Pt = 0.0018

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	16-04-2020	21-10-2020	12-10-2023				Date	16-04-2020	21-10-2020	12-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -95.534 - 12.828 = -108.362 kN-m
My1 = My - abs(Mxy) = -91.193 - 0.834 = -92.027 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -95.534 - 1.804 = -97.338 kN-m
My2 = My-abs(Mxy^2/Mx) = -91.193 - 0.007 = -91.186 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -108.362 kN-m
Myd = -92.027 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 108.362 kN-m
Φ*Kn = Mu/bd^2 deff. = 317 mm
Mu/bd^2 = 108.362X10^6/(1000X317^2) = 1.078 N/mm^2 < 6.9 N/mm^2
Pt = 0.0029 Reqd. Ast = 0.0029X1000X317/100 = 9.19 cm^2
Provided 16 mm @ 200 C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 O.K.
Total Provide Pt = 0.0032

ALONG LOCAL Y - DIRECTION :

Mu = 92.027 kN-m
Φ*Kn = Mu/bd^2 deff. = 301 mm
Mu/bd^2 = 92.027X10^6/(1000X301^2) = 1.016 N/mm^2 < 6.9 N/mm^2
Pt = 0.0027 Reqd. Ast = 0.0027X1000X301/100 = 8.13 cm^2
Provided 16 mm @ 200 C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 O.K.
Total Provide Pt = 0.0033

SHEAR CHECK

(Staad Group : SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = -0.63 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 837 , L/C No. - 305)*
Vu = 199.71 kN

Provided Pt = 0.0032
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0032X199.71X10^3X317/108.362/10^6)X1000X317/10^3
= 310.138 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0032)X1000X317/10^3
= 317.308 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X317/10^3
= 543.865 kN
HENCE, Vc = 310.138 kN
Φt*Vc = 0.75X310.138 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 232.604 kN > 199.71T , Hence O.K.

ALONG LOCAL Y - DIRECTION :

SQy = -0.635 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 837 , L/C No. - 305)*
Vu = 191.135 kN

Pt provided = 0.0033
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0033X191.135X10^3X301/92.027/10^3)X1000X301/10^3
= 295.475 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0033)X1000X301/10^3
= 301.805 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X301/10^3
= 516.415 kN
HENCE, Vc = 295.475 kN
Φt*Vc = 0.75X295.475 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 221.606 kN > 191.135T , Hence O.K.

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	16-04-2020	21-10-2020	12-10-2023	Date	16-04-2020	21-10-2020	12-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL	Prepared By	APL	APL	APL
Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

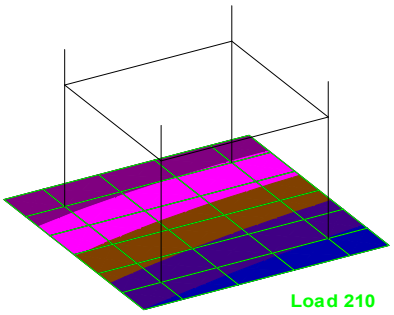
7.3.4 Design of combined footing mkd. CF1, CF2 & CF3

For Calculation of Base pressure and design, Refer STAAD File: 3210-CNV-001_TR1 FDN MKD_Combined.std

Staad Supoort Nodes are 5 & 6

Finished Floor level	F.F.L	=	0.3	m	(Corresponds to RL +912.750m)
Finished Ground level	F.G.L	=	0.0	m	(Corresponds to RL +912.250m)
Bottom of foundation	B.O.F	=	-1.40	m	(Corresponds to RL +910.000m)
Top of pedestal (staad support)	T.O.F	=	0.35	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	3.800	m	
Width of footing	Lx	=	5.000	m	
Thickness of footing	D	=	0.40	m	
Depth of pedestal	Bx	=	0.45	m	
Width of pedestal	Bz	=	0.45	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	525.2	KN/m^2	
Soil load on footing in staad		=	18	KN/m^2	staad input

Base Pressure
Y
kN/m2
<= -0.812
8.5
17.8
27.1
36.4
45.7
55.1
64.4
73.7
83
92.3
102
111
120
130
139
>= 148



1) Check for Gross Base Pressure :

Maximum Base Pressure	=	49.0	KN/m^2	Load Case 204
Minimum Base Pressure	=	-2.1	KN/m^2	Load Case 207

2) Design of foundation Raft :

Slab Thickness =	400	mm	Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Width =	1000	mm	Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Clear Cover =	75	mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Compressive Strength of Concrete , fc' =	35	N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

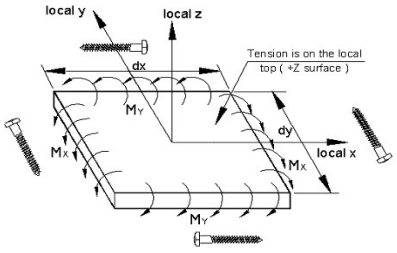
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : MX,MY)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	26.578	kN-m	Corresponding Mxy	-9.938	kN-m	*(Plate No. - 861 , L/C No. - 309)*
Min. Mx	-95.534	kN-m	Corresponding Mxy	12.828	kN-m	*(Plate No. - 861 , L/C No. - 304)*
Max. My	29.847	kN-m	Corresponding Mxy	-7.141	kN-m	*(Plate No. - 864 , L/C No. - 309)*
Min. My	-91.193	kN-m	Corresponding Mxy	-0.834	kN-m	*(Plate No. - 880 , L/C No. - 304)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	26.578 + 9.938 =	36.516 kN-m
My1 =	My + abs(Mxy) =	29.847 + 7.141 =	36.988 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	26.578 + 3.309 =	29.887 kN-m
My2 =	My+abs(Mxy^2/Mx) =	29.847 + 4.631 =	34.478 kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 36.516 kN-m
Myd = 36.988 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 36.516 kN-m
Φ*Kn = Mu/bd^2 deff. = 319 mm
Mu/bd^2 = 36.516X10^6/(1000X319^2) = 0.359 N/mm^2 < 6.9 N/mm^2
Pt = 0.0010 Reqd. Ast = 0.001X1000X319/100 = 3.19 cm^2
Provided 12 mm @ 200 C/C (Provide Ast = 5.65)cm^2

Total Provide Ast = 5.65 cm^2 O.K.
Total Provide Pt = 0.0018

ALONG LOCAL Y - DIRECTION :

Mu = 36.988 kN-m
Φ*Kn = Mu/bd^2 deff. = 307 mm
Mu/bd^2 = 36.988X10^6/(1000X307^2) = 0.363 N/mm^2 < 6.9 N/mm^2
Pt = 0.0010 Reqd. Ast = 0.001X1000X319/100 = 3.07 cm^2
Provided 12 mm @ 200 C/C (Provide Ast = 5.65)cm^2

Total Provide Ast = 5.65 cm^2 O.K.
Total Provide Pt = 0.0018

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	16-04-2020	21-10-2020	12-10-2023	Date	16-04-2020	21-10-2020	12-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL	Prepared By	APL	APL	APL
Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -95.534 - 12.828 = -108.362 kN-m
My1 = My - abs(Mxy) = -91.193 - 0.834 = -92.027 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -95.534 - 1.804 = -97.338 kN-m
My2 = My-abs(Mxy^2/Mx) = -91.193 - 0.007 = -91.186 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -108.362 kN-m
Myd = -92.027 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 108.362 kN-m
Φ*Kn = Mu/bd^2 deff. = 317 mm
Mu/bd^2 = 108.362X10^6/(1000X317^2) = 1.078 N/mm^2 < 6.9 N/mm^2
Pt = 0.0029 Reqd. Ast = 0.0029X1000X317/100 = 9.19 cm^2
Provided 16 mm @ 200 C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 O.K.
Total Provide Pt = 0.0032

ALONG LOCAL Y - DIRECTION :

Mu = 92.027 kN-m
Φ*Kn = Mu/bd^2 deff. = 301 mm
Mu/bd^2 = 92.027X10^6/(1000X301^2) = 1.016 N/mm^2 < 6.9 N/mm^2
Pt = 0.0027 Reqd. Ast = 0.0027X1000X301/100 = 8.13 cm^2
Provided 16 mm @ 200 C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 O.K.
Total Provide Pt = 0.0033

SHEAR CHECK

(Staad Group : SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = -0.63 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 837 , L/C No. - 305)*
Vu = 199.71 kN

Provided Pt = 0.0032
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0032X199.71X10^3X317/108.362/10^6)X1000X317/10^3
= 310.138 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0032)X1000X317/10^3
= 317.308 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X317/10^3
= 543.865 kN
HENCE, Vc = 310.138 kN
Φt*Vc = 0.75X310.138 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 232.604 kN **> 199.71T , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = -0.635 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 837 , L/C No. - 305)*
Vu = 191.135 kN

Pt provided = 0.0033
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0033X191.135X10^3X301/92.027/10^3)X1000X301/10^3
= 295.475 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0033)X1000X301/10^3
= 301.805 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X301/10^3
= 516.415 kN
HENCE, Vc = 295.475 kN
Φt*Vc = 0.75X295.475 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 221.606 kN **> 191.135T , Hence O.K.**

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	18-08-2020	21-10-2020	12-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT APL
Prepared By	APL	APL	SBY
Checked	SBY	SBY	PRI
Approved	PRI	PRI	

Outotec / L&T Revision No.:	0	1	2
Date	18-08-2020	21-10-2020	12-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT APL
Prepared By	APL	APL	SBY
Checked	SBY	SBY	PRI
Approved	PRI	PRI	

ATTACHMENT 1 **MECHANICAL GENERAL ARRANGEMENT DRAWINGS**

DRAWING NUMBER MD-216-3100-EG-ME-DED-V509-2002 REV A01

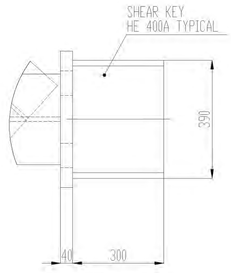
DESCRIPTION OF LOAD CASES	
LOAD CASE	LOAD TYPE
1	DEAD LOAD
2	LIVE LOAD
3	WIND LOAD +X DIRECTION
4	WIND LOAD -X DIRECTION
5	WIND LOAD +Z DIRECTION
6	WIND LOAD -Z DIRECTION
7	SEISMIC LOAD +X DIRECTION
8	SEISMIC LOAD -X DIRECTION
9	SEISMIC LOAD +Z DIRECTION
10	SEISMIC LOAD -Z DIRECTION

COLUMN	LOAD CASE	FX (kN)	FY (kN)	FZ (kN)
A1	1	-4	-216	0
	2	-8	-419	0
	3	138	388	25
	4	-137	-376	0
	5	-16	734	413
	6	15	-733	-7
	7	67	285	18
	8	-68	-285	0
	9	1	422	135
	10	-2	-423	0

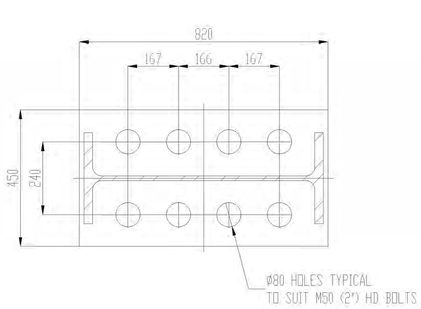
COLUMN	LOAD CASE	FX (kN)	FY (kN)	FZ (kN)
A2	1	4	-217	0
	2	8	-436	0
	3	140	-388	0
	4	-135	376	22
	5	-18	456	189
	6	20	-457	-6
	7	68	-285	0
	8	-67	285	18
	9	-1	422	135
	10	2	-422	0

COLUMN	LOAD CASE	FX (kN)	FY (kN)	FZ (kN)
A3	1	-4	-217	0
	2	-8	-419	0
	3	143	398	-19
	4	-149	-409	0
	5	14	-734	7
	6	-16	733	-412
	7	67	284	-18
	8	-68	-284	0
	9	-2	-422	0
	10	1	423	-135

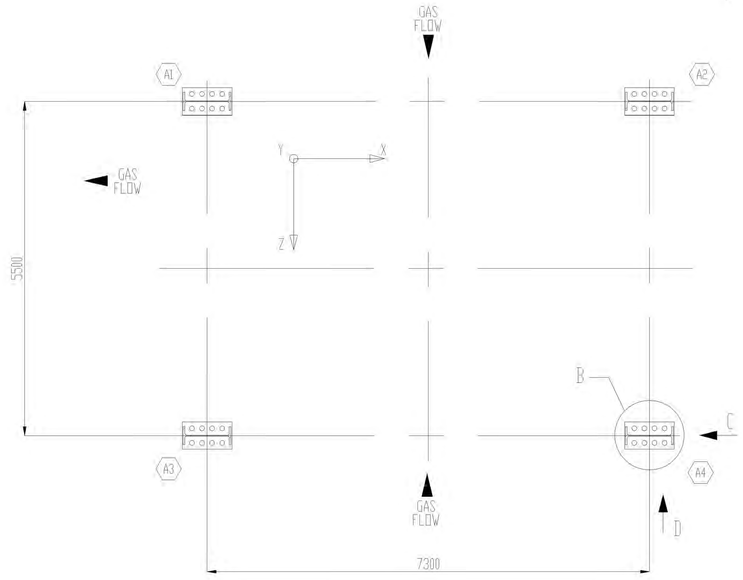
COLUMN	LOAD CASE	FX (kN)	FY (kN)	FZ (kN)
A4	1	4	-218	0
	2	8	-436	0
	3	145	-398	-6
	4	-146	409	-21
	5	19	-456	6
	6	-18	457	-190
	7	68	-284	0
	8	-67	284	-18
	9	2	-422	0
	10	-1	422	-135



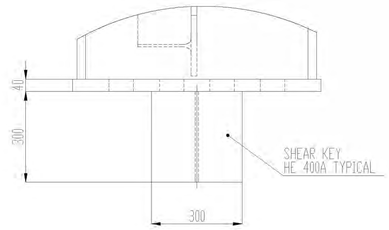
DETAIL C
SCALE 1 : 10



DETAIL B
SCALE 1 : 10



PLAN AT PA540x5000 BASEPLATES



DETAIL D
SCALE 1 : 10

GENERAL NOTES:
ANCHOR BOLTS TO ASTM F1554 SECTION 6.6.1 & TABLE 3
NUTS TO ASTM A563, GRADE A, HEAVY HEX WITH UNC-2B THREADS
WASHERS TO ASTM F436
ALTERNATE USE METRIC EQUIVALENTS

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A01	15/07/2020	ISSUED FOR INFORMATION		TKS	ABD	SBA	

FOWLEREX TECHNOLOGIES PTY. LTD.

DRAWING NUMBER					REVISION
10701-005					B
DATE	01/05/2020	DATE	02/07/2020	DATE	02/07/2020
DRAWN	AMc	CHECKED	AMc	APPROVED	AMc

LARSEN & TOUBRO DRAWING NUMBER: O19045_080_U_V_21419_002 — REVISION 0
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__CTB17__31007 — DOCUMENT ID: OU100682857 — REVISION 0

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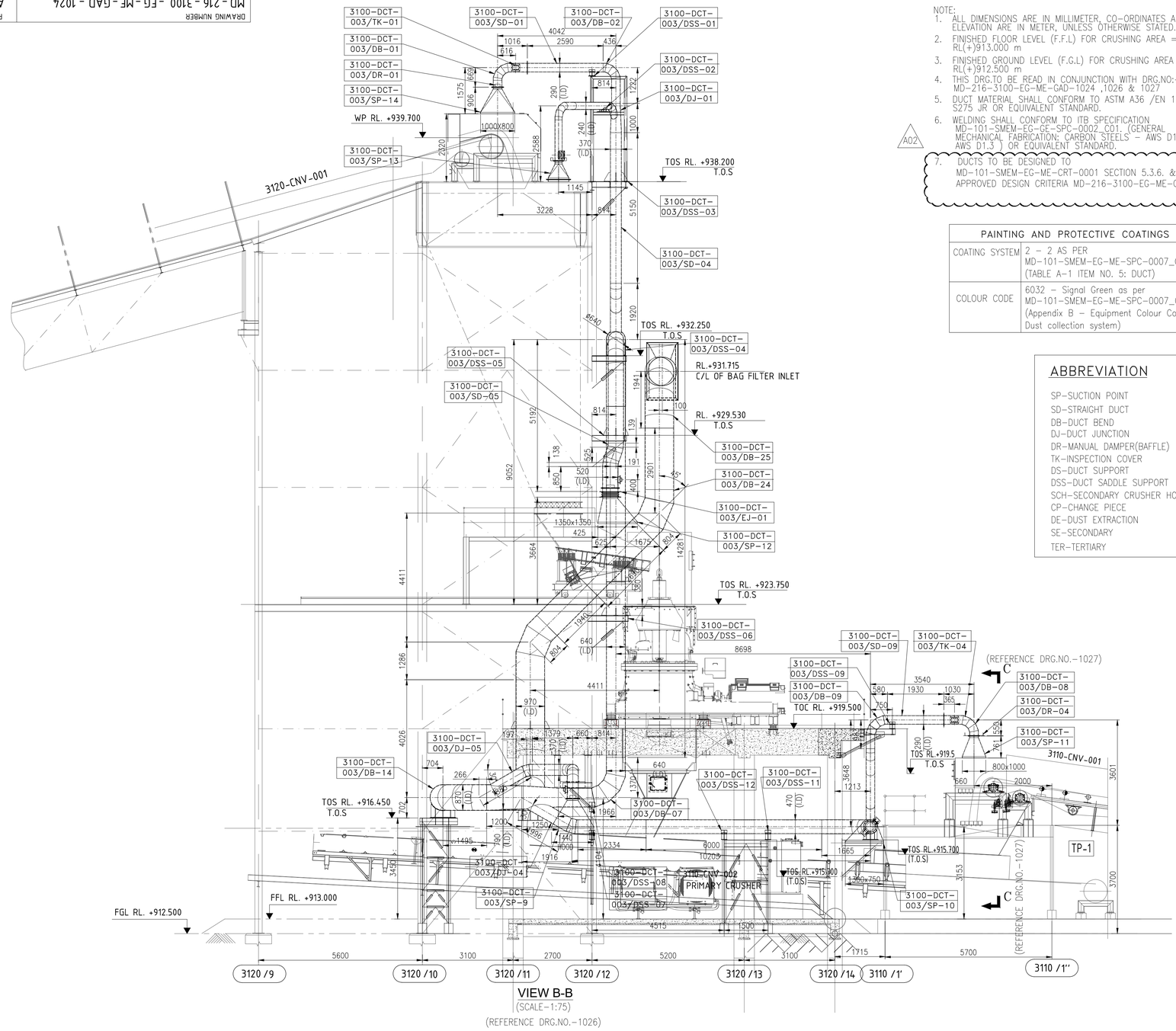
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DRAWN			MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	TKS	15/07/2020	DETAIL DRAWING
ENGINEER	ABD	15/07/2020	CRUSHING
APPROVED	SBA	15/07/2020	FDN LOAD DATA FOR SCREEN HOUSE BAG FILTER
EMPLOYER			
SCALE 1:50		A1	DRAWING NUMBER: MD-216-3100-EG-ME-DED-V509-2002 — REV: A01

DRAWING NUMBER: MD-216-3100-EG-ME-GAD-1024 | REV: A02



VIEW B-B
(SCALE-1:75)

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. THIS DRG.TO BE READ IN CONJUNCTION WITH DRG.NO:- MD-216-3100-EG-ME-GAD-1024 ,1026 & 1027
5. DUCT MATERIAL SHALL CONFORM TO ASTM A36 /EN 10025 S275 JR OR EQUIVALENT STANDARD.
6. WELDING SHALL CONFORM TO ITB SPECIFICATION MD-101-SMEM-EG-GE-SPC-0002_C01. (GENERAL MECHANICAL FABRICATION; CARBON STEELS - AWS D1.1, AWS D1.3) OR EQUIVALENT STANDARD.

A02
7. DUCTS TO BE DESIGNED TO MD-101-SMEM-EG-ME-CRT-0001 SECTION 5.3.6. & APPROVED DESIGN CRITERIA MD-216-3100-EG-ME-CRT-1003

PAINTING AND PROTECTIVE COATINGS	
COATING SYSTEM	2 - 2 AS PER MD-101-SMEM-EG-ME-SPC-0007_C01 (TABLE A-1 ITEM NO. 5: DUCT)
COLOUR CODE	6032 - Signal Green as per MD-101-SMEM-EG-ME-SPC-0007_C01 (Appendix B - Equipment Colour Coding: Dust collection system)

ABBREVIATION

SP-SUCTION POINT
SD-STRAIGHT DUCT
DB-DUCT BEND
DJ-DUCT JUNCTION
DR-MANUAL DAMPER(BAFFLE)
TK-INSPECTION COVER
DS-DUCT SUPPORT
DSS-DUCT SADDLE SUPPORT
SCH-SECONDARY CRUSHER HOUSE
CP-CHANGE PIECE
DE-DUST EXTRACTION
SE-SECONDARY
TER-TERTIARY

KEYPLAN

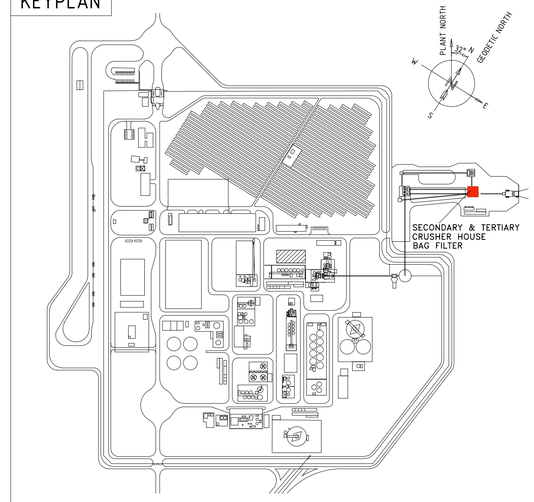


TABLE FOR SUPPORT MARKED & LOAD DATA

SL.NO.	Saddle Support Mark No.	Total Load IN Saddle(In MT.)	Remarks
1	3100-DCT-003/DSS-001	0.7	
2	3100-DCT-003/DSS-002	0.6	
3	3100-DCT-003/DSS-003	0.7	
4	3100-DCT-003/DSS-004	1.8	
5	3100-DCT-003/DSS-005	0.9	
6	3100-DCT-003/DSS-006	2.5	
7	3100-DCT-003/DSS-007	1.0	
8	3100-DCT-003/DSS-008	4.0	
9	3100-DCT-003/DSS-009	0.7	
10	3100-DCT-003/DSS-010	1.3	
11	3100-DCT-003/DSS-011	0.9	
12	3100-DCT-003/DSS-012	0.8	
13	3100-DCT-003/DSS-013	0.7	
14	3100-DCT-003/DSS-014	0.8	
15	3100-DCT-003/DSS-14A	8.0	
15	3100-DCT-003/DSS-14B	14.0	
15	3100-DCT-003/DSS-14C	20.0	
16	3100-DCT-003/DSS-015	5.0	
17	3100-DCT-003/DSS-016	0.8	
18	3100-DCT-003/DSS-017	4.0	
19	3100-DCT-003/DSS-018	8.0	
20	3100-DCT-003/DSS-019	6.0	
21	3100-DCT-003/DSS-020	8.0	

PAINTING AND PROTECTIVE COATINGS	
COATING SYSTEM	2 - 2 AS PER MD-101-SMEM-EG-ME-SPC-0007_C01 (TABLE A-1 ITEM NO. 5: DUCT)
COLOUR CODE	6032 - Signal Green as per MD-101-SMEM-EG-ME-SPC-0007_C01 (Appendix B - Equipment Colour Coding: Dust collection system)

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3100-EG-PR-PID-V509-2001	P&ID DUST EXTRACTION SYSTEM FOR SEC.& TER.CRUSHER HOUSE								
MD-216-3100-EG-ME-LAD-V509-2002	LAYOUT OF BAG FILTER & ID FAN								
MD-2016-3210-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHED ORE STORAGE								
MD-2016-3130-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING TER. CRUSHING								
MD-2016-3120-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING SEC. CRUSHING								
MD-2016-3110-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING PRIMARY CRUSHING	A02	23/07/2020	REVISED AS PER CUSTOMER COMMENT DATED-15/06/2020	TKS/ABW	SYR	ABD	SBA/JPD	
MD-316-3100-EG-ME-SDD-1001	SCHEMATIC DIAGRAM CRUSHING DUST EXTRACTION	A01	12/05/2020	ISSUED FOR APPROVAL	TKS/IRH	ABD	ABD	SBA/JPD	

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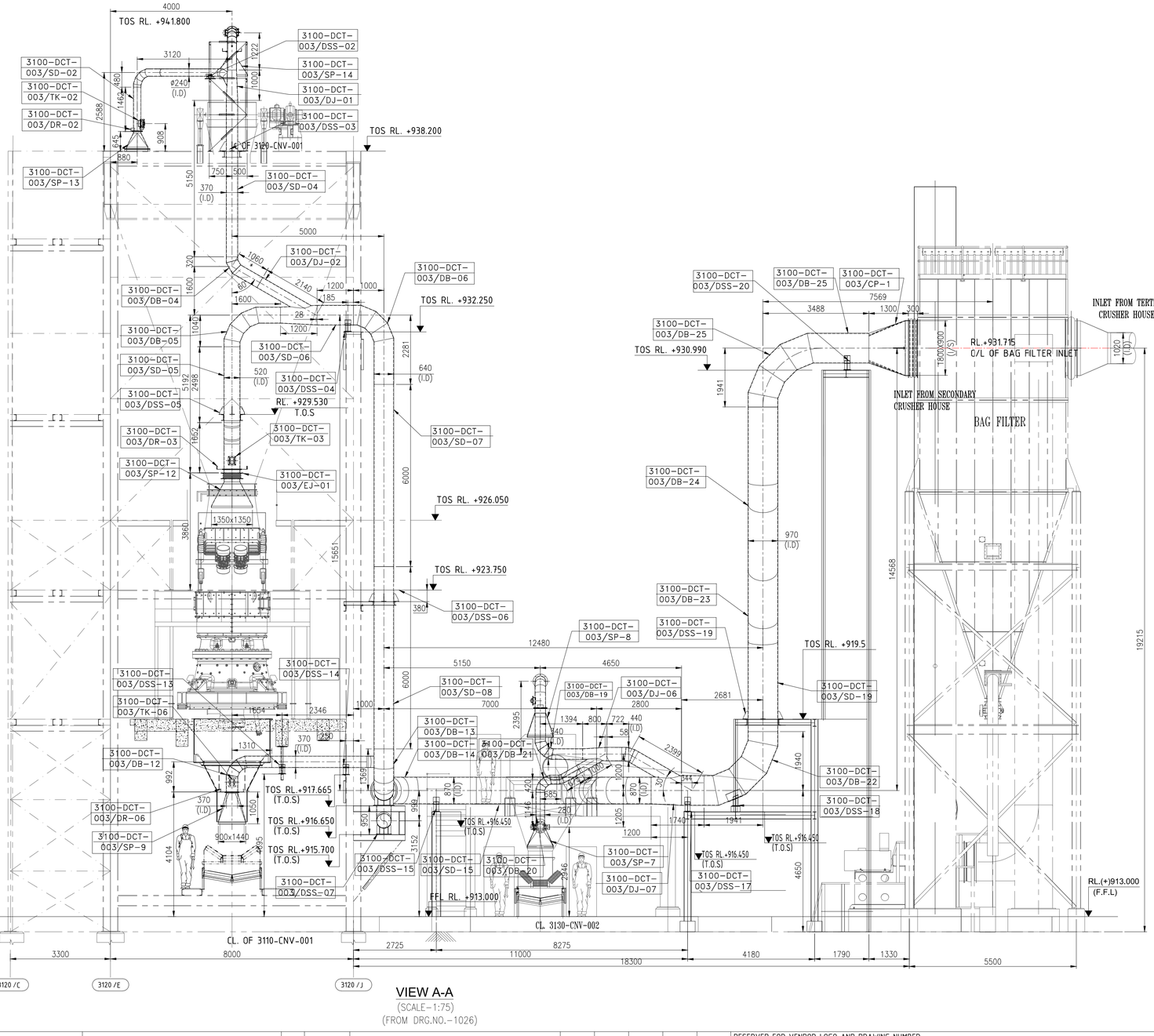
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CHECKED	ABD	10/05/2020
ENGINEER	ABD	10/05/2020
APPROVED	SBA/JPD	12/05/2020
EMPLOYER		
SCALE	1:75	A1

LARSEN & TOUBRO DRAWING NUMBER: 019045_080_U_G_21411_001 | REVISION: 1
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31012 | DOCUMENT ID: OU100672973 | REVISION: 1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
CRUSHING
DUCT-DUST EXTRACTION SYSTEM FOR SECONDARY CRUSHER HOUSE

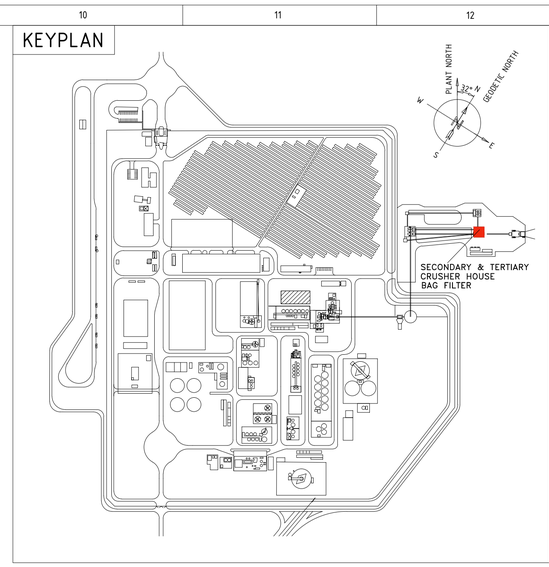
DRAWING NUMBER: MD-216-3100-EG-ME-GAD-1024 | REV: A02

DRAWING NUMBER MD-216-3100-EG-ME-GAD-1025 REV A02



VIEW A-A
(SCALE-1:75)
(FROM DRG.NO.-1026)

KEYPLAN



ABBREVIATION

SP-SUCTION POINT
SD-STRAIGHT DUCT
DB-DUCT BEND
DJ-DUCT JUNCTION
DR-MANUAL DAMPER(BAFFLE)
TK-INSPECTION COVER
DS-DUCT SUPPORT
DSS-DUCT SADDLE SUPPORT
SCH-SECONDARY CRUSHER HOUSE
CP-CHANGE PIECE
DE-DUST EXTRACTION
SE-SECONDARY
TER-TERTIARY

PAINTING AND PROTECTIVE COATINGS	
COATING SYSTEM	2 - 2 AS PER MD-101-SMEM-EG-ME-SPC-0007_C01 (TABLE A-1 ITEM NO. 5: DUCT)
COLOUR CODE	6032 - Signal Green as per MD-101-SMEM-EG-ME-SPC-0007_C01 (Appendix B - Equipment Colour Coding: Dust collection system)

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. THIS DRG.TO BE READ IN CONJUNCTION WITH DRG.NO:- MD-216-3100-EG-ME-GAD-1024 ,1026 & 1027
5. DUCT MATERIAL SHALL CONFORM TO ASTM A36 /EN 10025 S275 JR OR EQUIVALENT STANDARD.
6. WELDING SHALL CONFORM TO ITB SPECIFICATION MD-101-SMEM-EG-GE-SPC-0002_C01. (GENERAL MECHANICAL FABRICATION: CARBON STEELS - AWS D1.1, AWS D1.3) OR EQUIVALENT STANDARD.

A02

7. DUCTS TO BE DESIGNED TO MD-101-SMEM-EG-ME-CRT-0001 SECTION 5.3.6. & APPROVED DESIGN CRITERIA MD-216-3100-EG-ME-CRT-1003

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3100-EG-PR-PID-V509-2001	P&ID DUST EXTRACTION SYSTEM FOR SEC.& TER.CRUSHER HOUSE								
MD-316-3100-EG-ME-LAD-V509-2002	LAYOUT OF BAG FILTER & ID FAN								
MD-2016-3210-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHED ORE STORAGE								
MD-2016-3130-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING TER. CRUSHING								
MD-2016-3120-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING SEC. CRUSHING								
MD-2016-3110-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING PRIMARY CRUSHING	A02	23/07/2020	REVISED AS PER CUSTOMER COMMENT DATED-15/06/2020	TKS/ABW	SYR	ABD	SBA/JPD	
MD-316-3100-EG-ME-SDD-1001	SCHEMATIC DIAGRAM CRUSHING DUST EXTRACTION	A01	12/05/2020	ISSUED FOR APPROVAL	TKS/IRH	ABD	ABD	SBA/JPD	

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OUTOTEC DRAWING NUMBER: 900707 _GAU01 _ZC01 _MTL01 _31013 — DOCUMENT ID: OU100672974 — REVISION: 1

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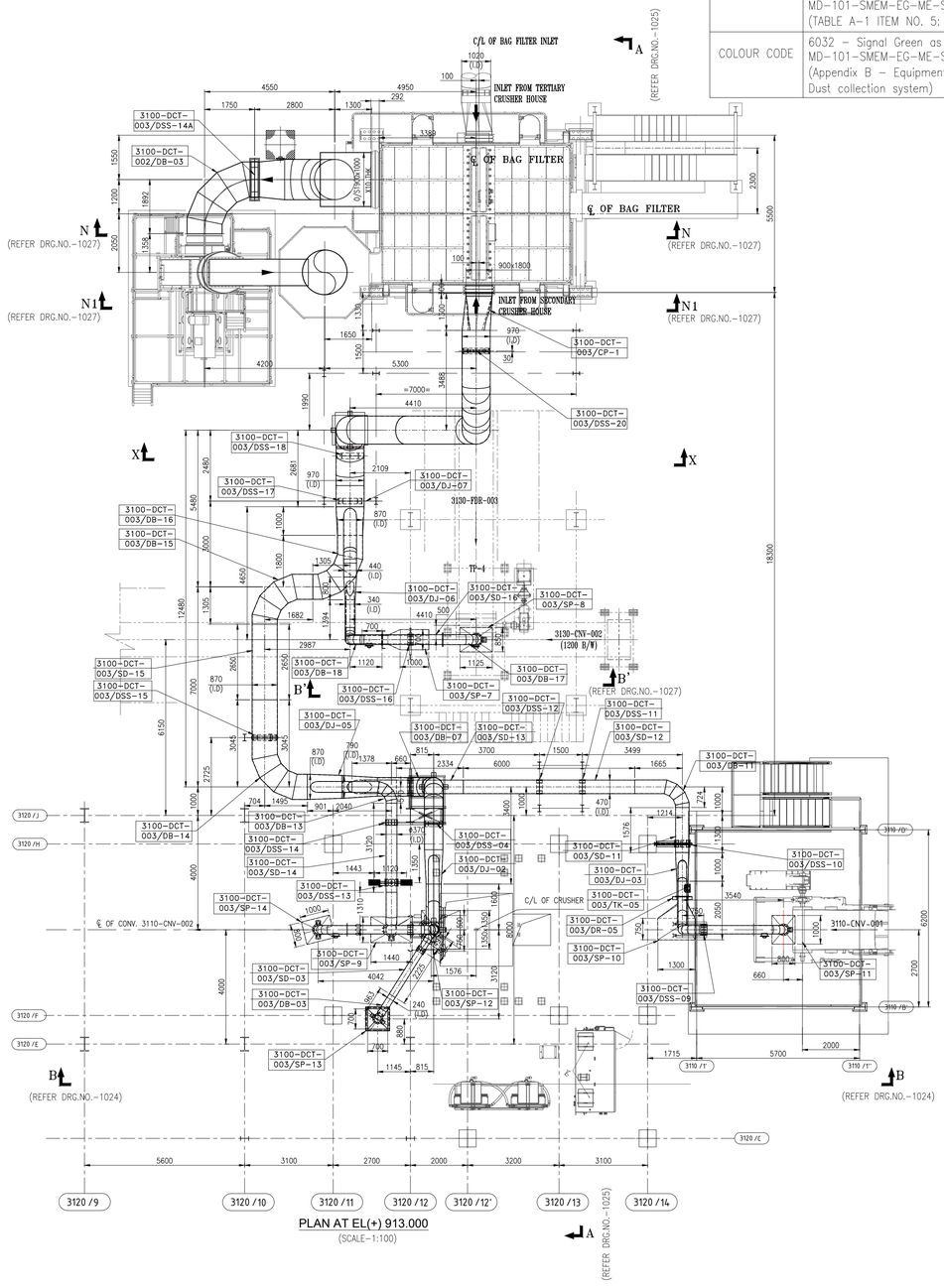
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CHECKED	ABD	10/05/2020	
ENGINEER	ABD	10/05/2020	
APPROVED	SBA/JPD	12/05/2020	
EMPLOYER			
SCALE	1:75	A1	

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
CRUSHING
DUCT-DUST EXTRACTION SYSTEM FOR SECONDARY CRUSHER HOUSE

DRAWING NUMBER: MD - 216 - 3100 - EG - ME - GAD - 1025 — REV: A02

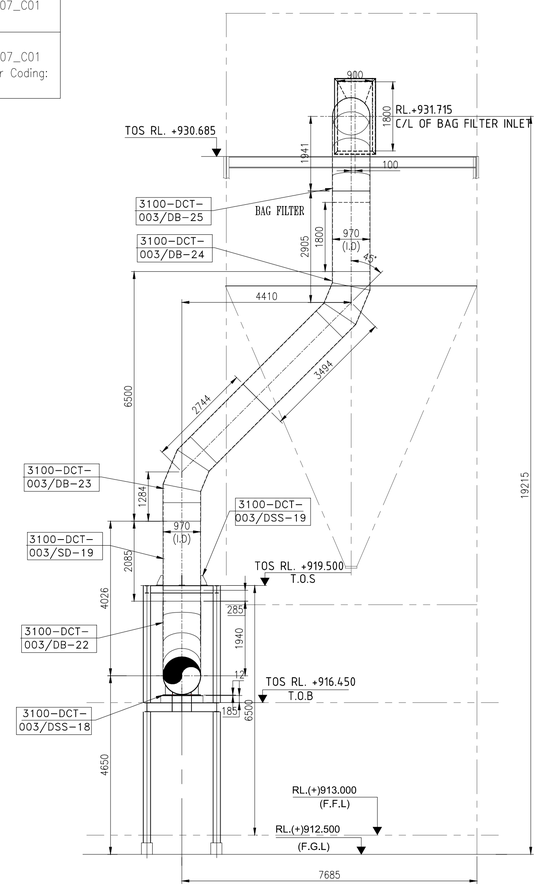
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MD-216-3100-EG-ME-GAD-1026	A02

PAINTING AND PROTECTIVE COATINGS	
COATING SYSTEM	2 – 2 AS PER MD-101-SMEM-EG-ME-SPC-0007_C01 (TABLE A-1 ITEM NO. 5: DUCT)
COLOUR CODE	6032 – Signal Green as per MD-101-SMEM-EG-ME-SPC-0007_C01 (Appendix B – Equipment Colour Coding: Dust collection system)



PLAN AT EL(+) 913.000

(SCALE-1:100)



VIEW X-X

(SCALE-1:75)

NOTE:

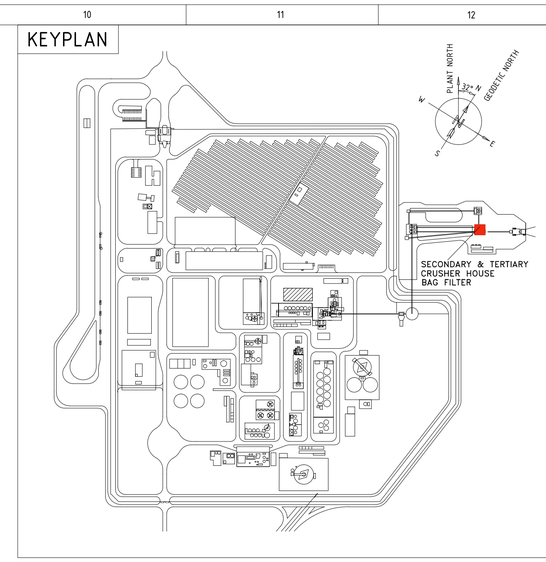
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3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
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5. DUCT MATERIAL SHALL CONFORM TO ASTM A36 /EN 10025 S275 JR OR EQUIVALENT STANDARD.
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7. DUCTS TO BE DESIGNED TO MD-101-SMEM-EG-ME-CRT-0001 SECTION 5.3.6. & APPROVED DESIGN CRITERIA MD-216-3100-EG-ME-CRT-1003

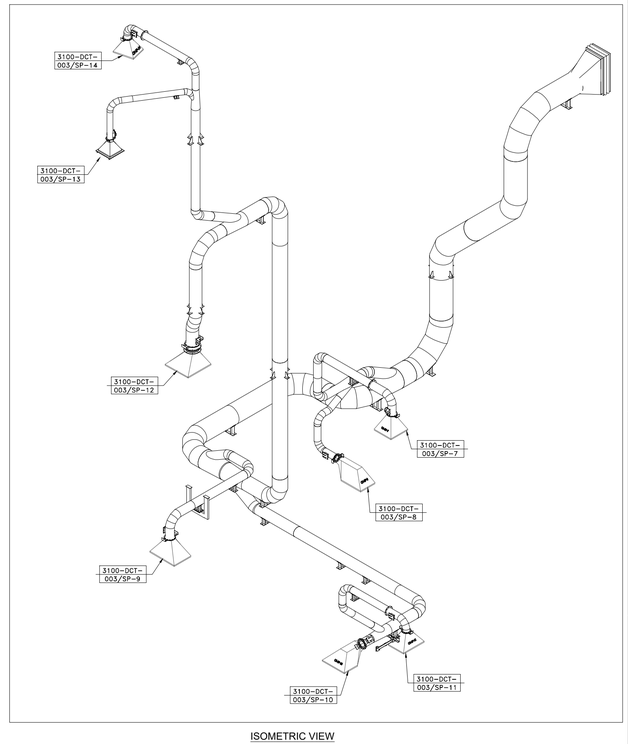
A02

ABBREVIATION

SP	SUCTION POINT
SD	STRAIGHT DUCT
DB	DUCT BEND
DJ	DUCT JUNCTION
DR	MANUAL DAMPER(BAFFLE)
TK	INSPECTION COVER
DS	DUCT SUPPORT
DSS	DUCT SADDLE SUPPORT
SCH	SECONDARY CRUSHER HOUSE
CP	CHANGE PIECE
DE	DUST EXTRACTION
SE	SECONDARY
TER	TERTIARY

KEYPLAN





ISOMETRIC VIEW

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3100-EG-PR-PID-V509-2001	P&ID DUST EXTRACTION SYSTEM FOR SEC.& TER.CRUSHER HOUSE								
MD-216-3100-EG-ME-LAD-V509-2002	LAYOUT OF BAG FILTER & ID FAN								
MD-2016-3210-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHED ORE STORAGE								
MD-2016-3130-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING TER. CRUSHING								
MD-2016-3120-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING SEC. CRUSHING								
MD-2016-3110-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING PRIMARY CRUSHING	A02	23/07/2020	REVISED AS PER CUSTOMER COMMENT DATED-15/06/2020	TKS/ABW	SYR	ABD	SBA/JPD	
MD-316-3100-EG-ME-SDD-1001	SCHEMATIC DIAGRAM CRUSHING DUST EXTRACTION	A01	12/05/2020	ISSUED FOR APPROVAL	TKS/IRH	ABD	ABD	SBA/JPD	

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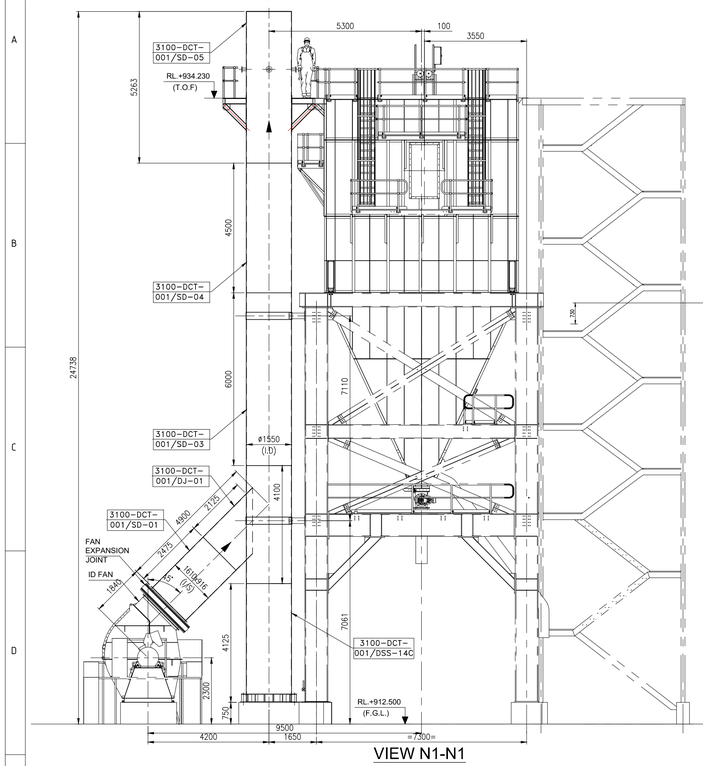
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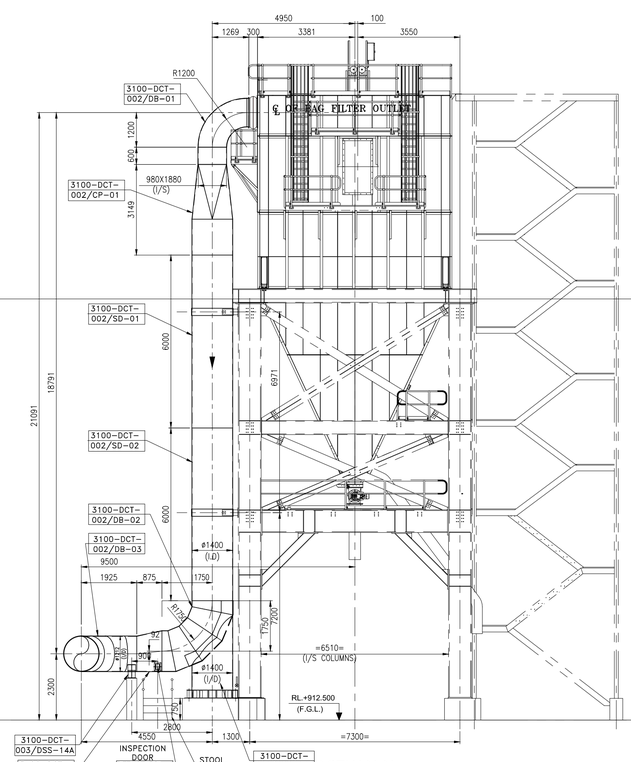
LARSEN & TOUBRO DRAWING NUMBER			REVISION
O19045_080_U_G_21411_003			1
OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
900707 _GAU01 _ZC01 _MTL01 _31014		OU100672975	1
DRAWN	TKS/IRH	08/05/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	ABD	10/05/2020	GENERAL ARRANGEMENT
ENGINEER	ABD	10/05/2020	CRUSHING
APPROVED	SBA/JPD	12/05/2020	DUCT-DUST EXTRACTION SYSTEM FOR SECONDARY CRUSHER HOUSE
EMPLOYER			
SCALE 1:100	A1		DRAWING NUMBER: MD-216-3100-EG-ME-GAD-1026 REV: A02

DRAWING NUMBER: MD-216-3100-EG-ME-GAD-1027 | REV: A02



VIEW N1-N1

SCALE 1:100
(FROM DRG.NO.-1026)



VIEW N-N

SCALE 1:100
(FROM DRG.NO.-1026)

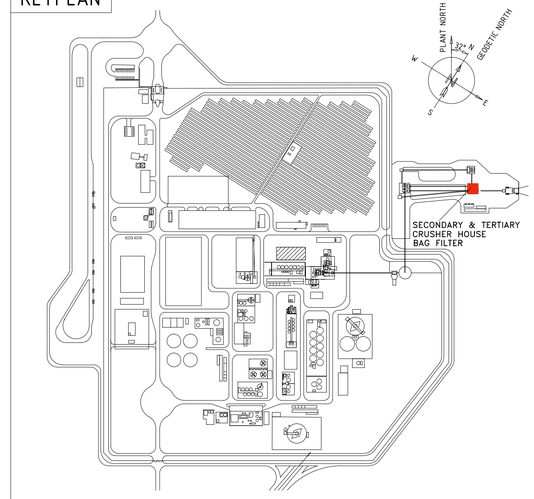
ABBREVIATION

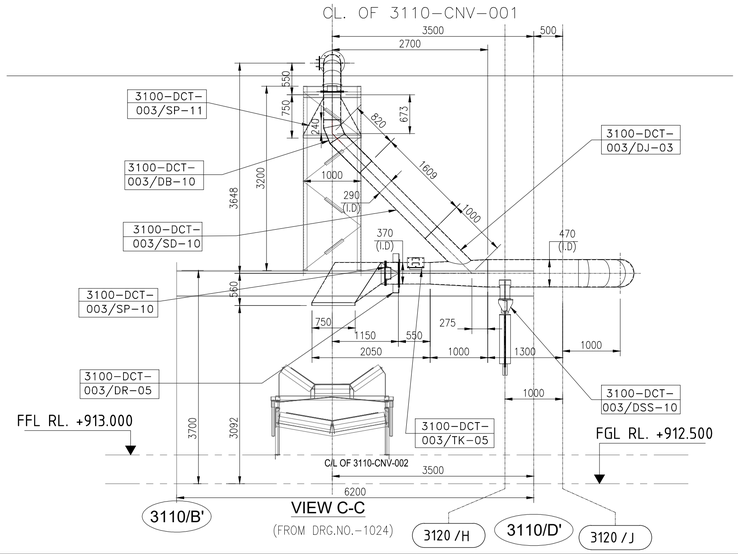
SP	SUCTION POINT
SD	STRAIGHT DUCT
DB	DUCT BEND
DJ	DUCT JUNCTION
DR	MANUAL DAMPER(BAFFLE)
TK	INSPECTION COVER
DS	DUCT SUPPORT
DSS	DUCT SADDLE SUPPORT
SCH	SECONDARY CRUSHER HOUSE
CP	CHANGE PIECE
DE	DUST EXTRACTION
SE	SECONDARY
TER	TERTIARY

PAINTING AND PROTECTIVE COATINGS

COATING SYSTEM	2 - 2 AS PER MD-101-SMEM-EG-ME-SPC-0007_C01 (TABLE A-1 ITEM NO. 5: DUCT)
COLOUR CODE	6032 - Signal Green as per MD-101-SMEM-EG-ME-SPC-0007_C01 (Appendix B - Equipment Colour Coding: Dust collection system)

KEYPLAN





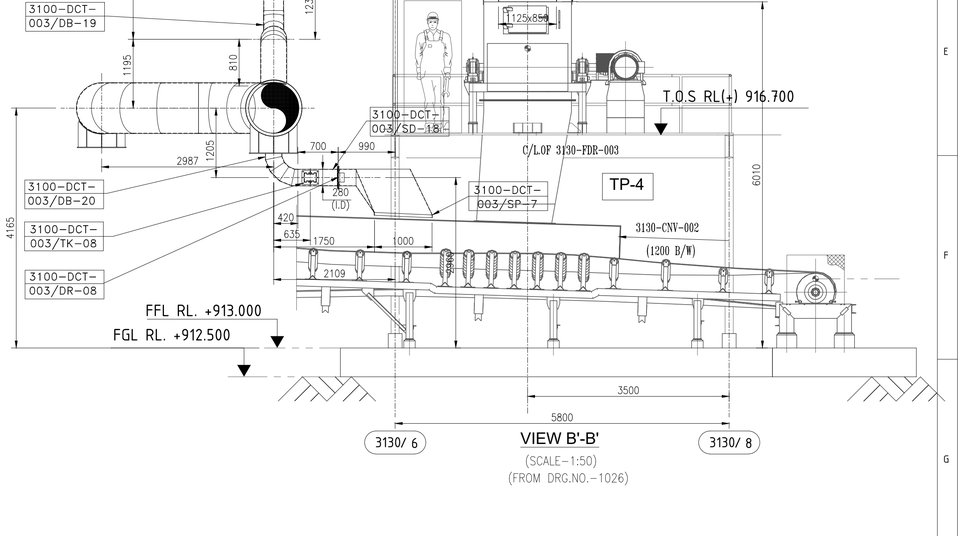
VIEW C-C

(FROM DRG.NO.-1024)

NOTE:

1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. THIS DRG.TO BE READ IN CONJUNCTION WITH DRG.NO:- MD-216-3100-EG-ME-GAD-1024 ,1026 & 1027
5. DUCT MATERIAL SHALL CONFORM TO ASTM A36 /EN 10025 S275 JR OR EQUIVALENT STANDARD.
6. WELDING SHALL CONFORM TO ITB SPECIFICATION MD-101-SMEM-EG-GE-SPC-0002_C01. (GENERAL MECHANICAL FABRICATION: CARBON STEELS - AWS D1.1, AWS D1.3) OR EQUIVALENT STANDARD.
7. DUCTS TO BE DESIGNED TO MD-101-SMEM-EG-ME-CRT-0001 SECTION 5.3.6. & APPROVED DESIGN CRITERIA MD-216-3100-EG-ME-CRT-1003

A02



VIEW B'-B'

(SCALE-1:50)
(FROM DRG.NO.-1026)

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3100-EG-PR-PID-V509-2001	P&ID DUST EXTRACTION SYSTEM FOR SEC.& TER.CRUSHER HOUSE								
MD-216-3100-EG-ME-LAD-V509-2002	LAYOUT OF BAG FILTER & ID FAN								
MD-2016-3210-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHED ORE STORAGE								
MD-2016-3130-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING TER. CRUSHING								
MD-2016-3120-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING SEC. CRUSHING								
MD-2016-3110-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING PRIMARY CRUSHING	A02	23/07/2020	REVISED AS PER CUSTOMER COMMENT DATED-15/06/2020	TKS/ABW	SYR	ABD	SBA/JPD	
MD-316-3100-EG-ME-SDD-1001	SCHEMATIC DIAGRAM CRUSHING DUST EXTRACTION	A01	12/05/2020	ISSUED FOR APPROVAL	TKS/IRH	ABD	ABD	SBA/JPD	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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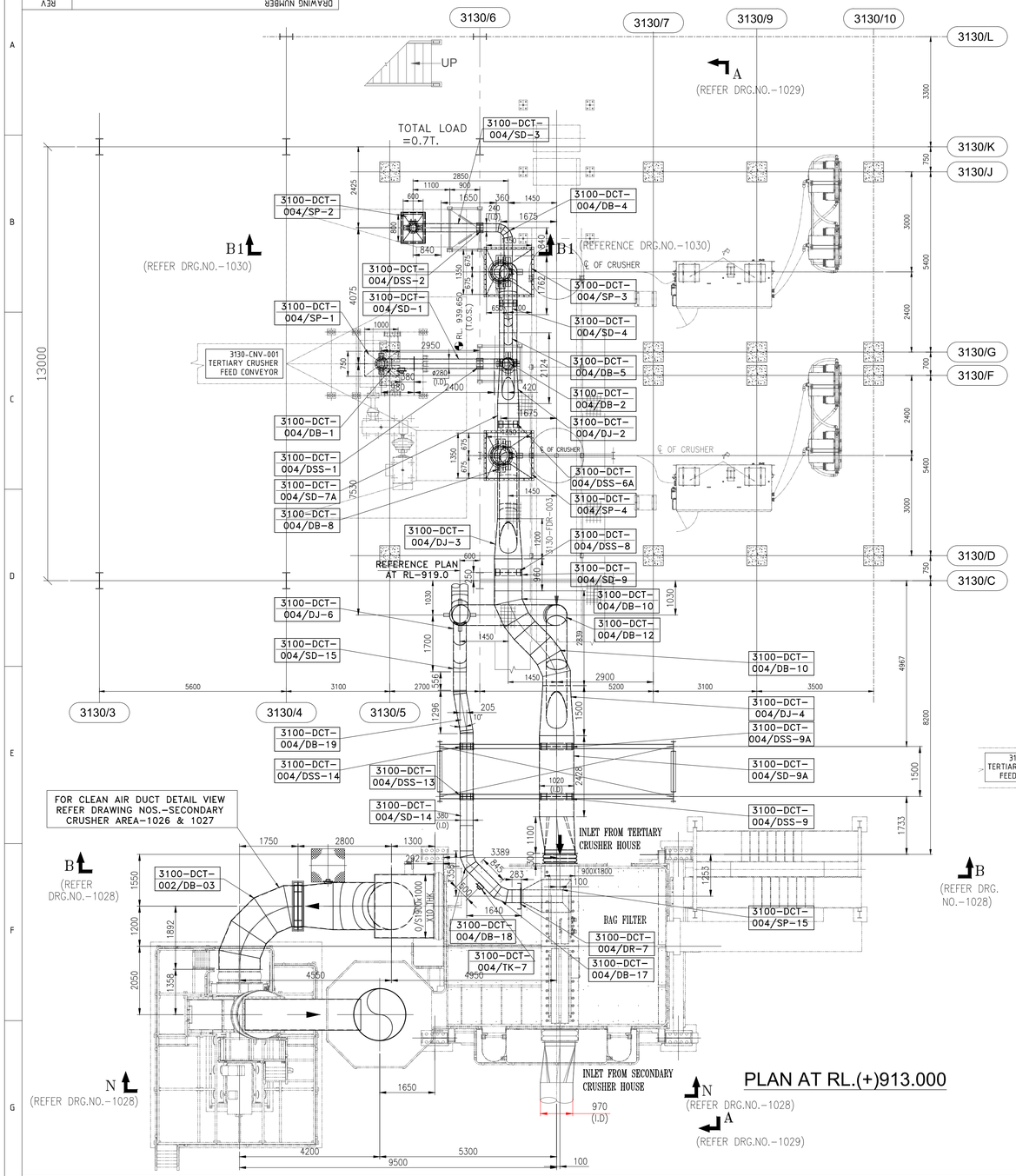
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CHECKED	ABD	10/05/2020	
ENGINEER	ABD	10/05/2020	
APPROVED	SBA/JPD	12/05/2020	
EMPLOYER			
SCALE	1:100, 50	A1	

LARSEN & TOUBRO DRAWING NUMBER: 019045_080_U_G_21411_004 | REVISION: 1
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31015 | DOCUMENT ID: OU100672976 | REVISION: 1

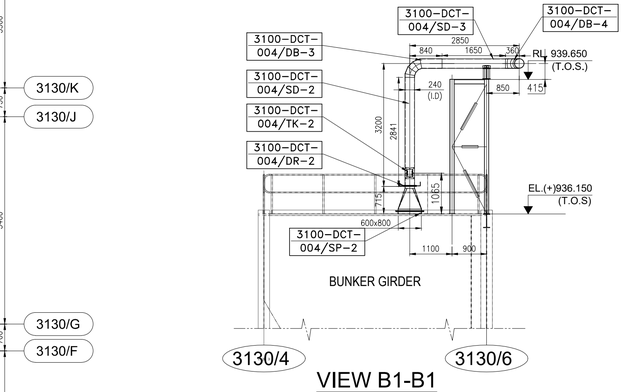
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
CRUSHING
DUCT-DUST EXTRACTION SYSTEM FOR SECONDARY CRUSHER HOUSE

DRAWING NUMBER: MD-216-3100-EG-ME-GAD-1027 | REV: A02

DRAWING NUMBER: MD-216-3100-EG-ME-GAD-1030 | REV: A02

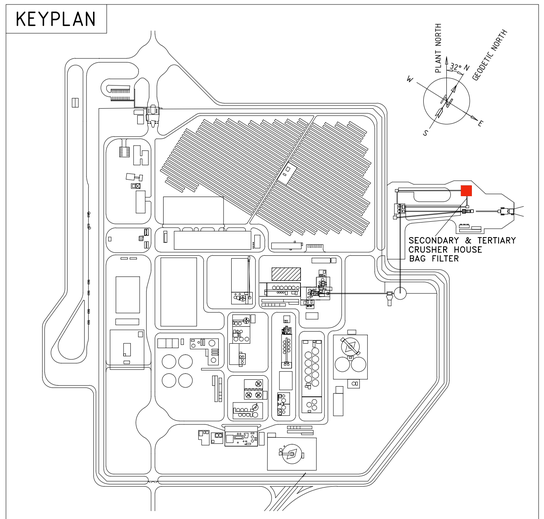


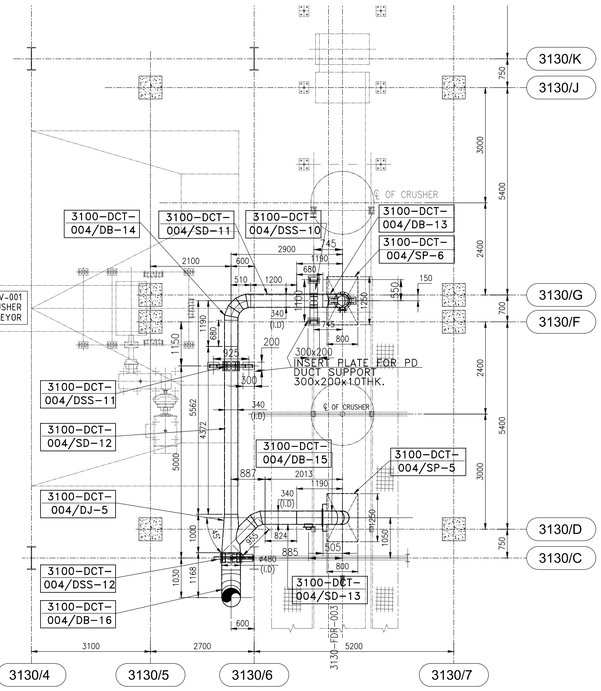
PLAN AT RL.(+)913.000



VIEW B1-B1
(FROM DRG.NO.-1030)

KEYPLAN





PLAN AT RL.(+)919.000

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. THIS DRG.TO BE READ IN CONJUNCTION WITH DRG.NO:- MD-216-3100-EG-ME-GAD-1028,1029
5. DUCT MATERIAL SHALL CONFORM TO ASTM A36 /EN 10025 S275 JR OR EQUIVALENT STANDARD.
6. WELDING SHALL CONFORM TO ITB SPECIFICATION MD-101-SMEM-EG-GE-SPC-0002_C01. (GENERAL MECHANICAL FABRICATION: CARBON STEELS - AWS D1.1, AWS D1.3) OR EQUIVALENT STANDARD.
7. DUCTS TO BE DESIGNED TO MD-101-SMEM-EG-ME-CRT-0001 SECTION 5.3.6. & APPROVED DESIGN CRITERIA MD-216-3100-EG-ME-CRT-1003

A02

PAINTING AND PROTECTIVE COATINGS	
COATING SYSTEM	2 - 2 AS PER MD-101-SMEM-EG-ME-SPC-0007_C01 (TABLE A-1 ITEM NO. 5: DUCT)
COLOUR CODE	6032 - Signal Green as per MD-101-SMEM-EG-ME-SPC-0007_C01 (Appendix B - Equipment Colour Coding: Dust collection system)

ABBREVIATION

SP-SUCTION POINT
SD-STRAIGHT DUCT
DB-DUCT BEND
DJ-DUCT JUNCTION
DR-MANUAL DAMPER(BAFFLE)
TK-INSPECTION COVER
DS-DUCT SUPPORT
DSS-DUCT SADDLE SUPPORT
SCH-SECONDARY CRUSHER HOUSE
CP-CHANGE PIECE
DE-DUST EXTRACTION
SYS-SYSTEM
TER-TERTIARY

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3100-EG-PR-PID-V509-2001	P&ID DUST EXTRACTION SYSTEM FOR SEC.& TER.CRUSHER HOUSE								
MD-216-3100-EG-ME-LAD-V509-2002	LAYOUT OF BAG FILTER & ID FAN								
MD-2016-3210-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHED ORE STORAGE								
MD-2016-3130-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING TER. CRUSHING								
MD-2016-3120-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING SEC. CRUSHING								
MD-2016-3110-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING PRIMARY CRUSHING	A02	17/07/2020	REVISED AS PER CUSTOMER COMMENT DATED-15/06/2020	TKS	SYR	ABD	SBA/JPD	
MD-316-3100-EG-ME-SDD-1001	SCHEMATIC DIAGRAM CRUSHING DUST EXTRACTION	A01	12/05/2020	ISSUED FOR APPROVAL	TKS	ABD	ABD	SBA/JPD	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER: O19045_080_U_G_21417_003 | REVISION: 1
OUTOTEC DRAWING NUMBER: 900707 _GAU01 _ZC01 _MTL01 _31018 | DOCUMENT ID: OU100672979 | REVISION: 1

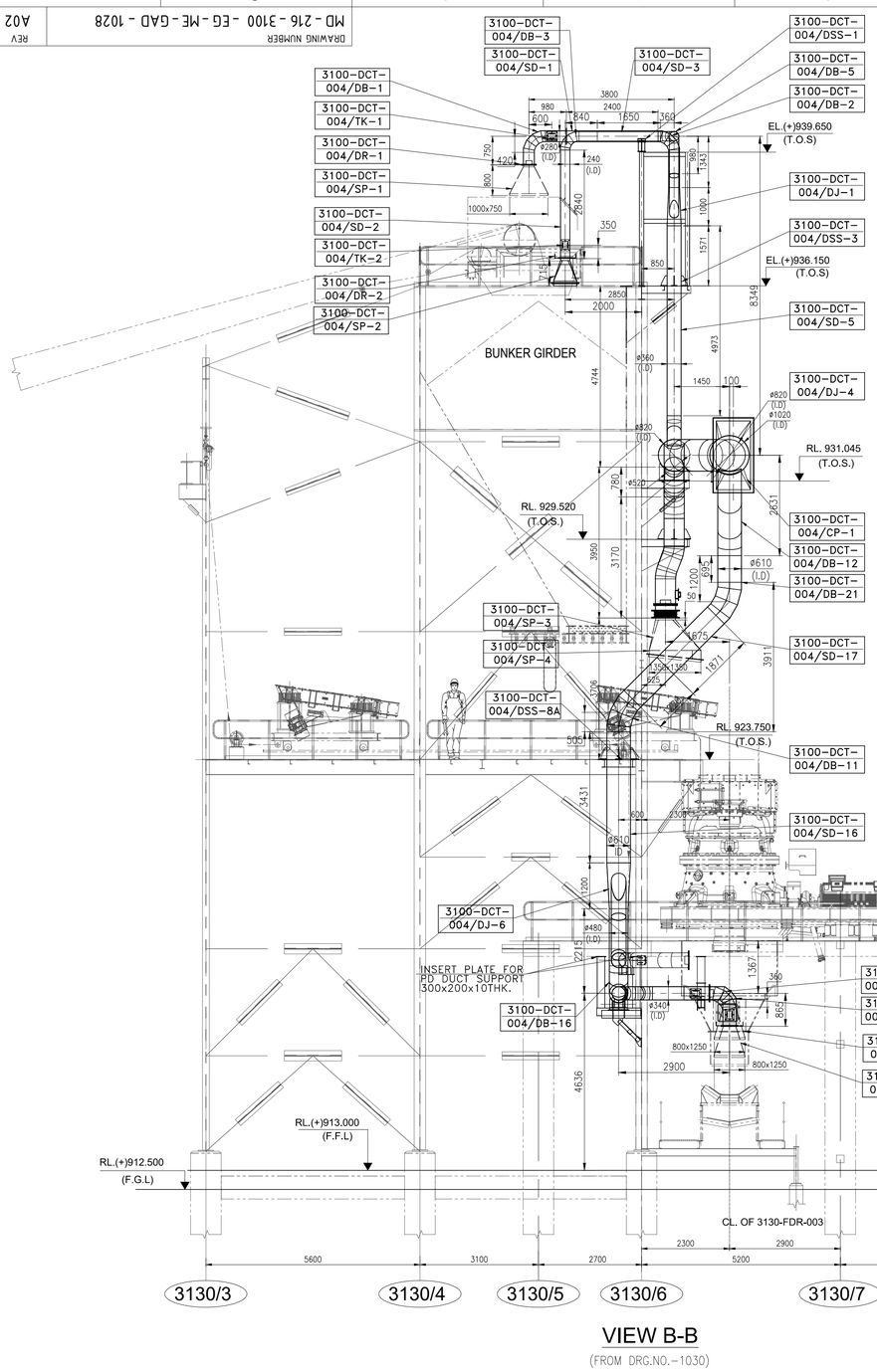
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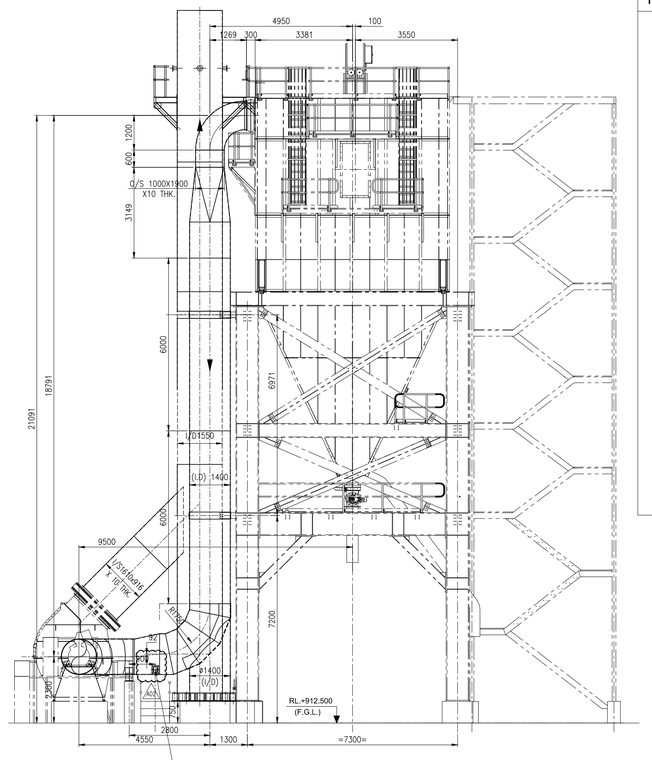
DRAWN	TKS	08/05/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	ABD	10/05/2020	GENERAL ARRANGEMENT
ENGINEER	ABD	10/05/2020	CRUSHING
APPROVED	SBA/JPD	12/05/2020	DUCT-DUST EXTRACTION SYSTEM FOR TERTIARY CRUSHER HOUSE
EMPLOYER			
SCALE	1:75	A1	DRAWING NUMBER: MD-216-3100-EG-ME-GAD-1030 | REV: A02

DRAWING NUMBER: MD-216-3100-EG-ME-GAD-1028 | REV: A02



VIEW B-B

(FROM DRG.NO.-1030)



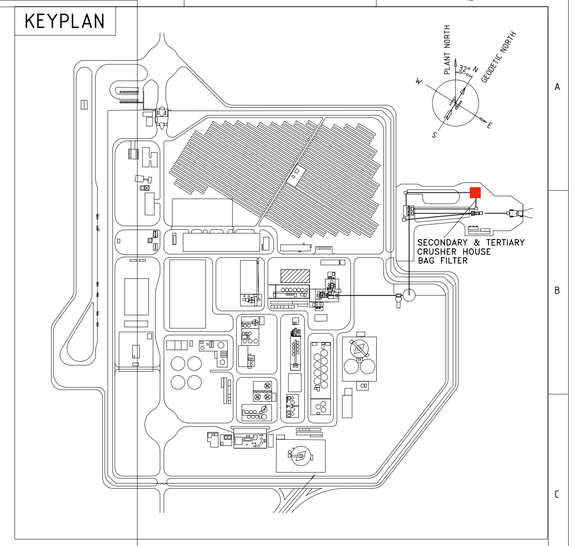
VIEW N-N

SCALE 1:100
(FROM DRG.NO.-1030)

TABLE FOR SUPPORT MARKED & LOAD DATA

SL.NO.	Saddle Support Mark No.	Total Load IN Saddle(In T.)	Remarks
1	3100-DCT-004/DSS-001	1.0	
2	3100-DCT-004/DSS-002	0.7	
3	3100-DCT-004/DSS-003	0.6	
4	3100-DCT-004/DSS-004	0.4	
5	3100-DCT-004/DSS-005	1.0	
6	3100-DCT-004/DSS-006	2.0	
7	3100-DCT-004/DSS-06A	2.0	
8	3100-DCT-004/DSS-007	0.6	
9	3100-DCT-004/DSS-008	5.5	
10	3100-DCT-004/DSS-8A	2.0	
11	3100-DCT-004/DSS-009	4.0	
12	3100-DCT-004/DSS-09A	8.0	
13	3100-DCT-004/DSS-010	0.5	
14	3100-DCT-004/DSS-011	0.7	
15	3100-DCT-004/DSS-012	2.2	
16	3100-DCT-004/DSS-013	0.8	
17	3100-DCT-004/DSS-014	1.0	

KEYPLAN



NOTE:

1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
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3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. THIS DRG.TO BE READ IN CONJUNCTION WITH DRG.NO:- MD-216-3100-EG-ME-GAD-1029,1030
5. DUCT MATERIAL SHALL CONFORM TO ASTM A36 /EN 10025 S275 JR OR EQUIVALENT STANDARD.
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7. DUCTS TO BE DESIGNED TO MD-101-SMEM-EG-ME-CRT-0001 SECTION 5.3.6. & APPROVED DESIGN CRITERIA MD-216-3100-EG-ME-CRT-1003

A02

PAINTING AND PROTECTIVE COATINGS	
COATING SYSTEM	2 - 2 AS PER MD-101-SMEM-EG-ME-SPC-0007_C01 (TABLE A-1 ITEM NO. 5: DUCT)
COLOUR CODE	6032 - Signal Green as per MD-101-SMEM-EG-ME-SPC-0007_C01 (Appendix B - Equipment Colour Coding: Dust collection system)

ABBREVIATION

SP-SUCTION POINT
SD-STRAIGHT DUCT
DB-DUCT BEND
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DR-MANUAL DAMPER(BAFFLE)
TK-INSPECTION COVER
DS-DUCT SUPPORT
DSS-DUCT SADDLE SUPPORT
SCH-SECONDARY CRUSHER HOUSE
CP-CHANGE PIECE
DE-DUST EXTRACTION
SYS-SYSTEM
TER-TIERTIARY

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3100-EG-PR-PID-V509-2001	P&ID DUST EXTRACTION SYSTEM FOR SEC.& TER.CRUSHER HOUSE								
MD-216-3100-EG-ME-LAD-V509-2002	LAYOUT OF BAG FILTER & ID FAN								
MD-2016-3210-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHED ORE STORAGE								
MD-2016-3130-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING TER. CRUSHING								
MD-2016-3120-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING SEC. CRUSHING								
MD-2016-3110-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING PRIMARY CRUSHING	A02	17/07/2020	REVISED AS PER CUSTOMER COMMENT DATED-15/06/2020	TKS	SYR	ABD	SBA/JPD	
MD-316-3100-EG-ME-SDD-1001	SCHEMATIC DIAGRAM CRUSHING DUST EXTRACTION	A01	12/05/2020	ISSUED FOR APPROVAL	TKS	ABD/TKS	ABD	SBA/JPD	

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OUTOTEC DRAWING NUMBER: 900707 _GAU01 _ZC01 _MTL01 _31016 | DOCUMENT ID: OU100672977 | REVISION: 1

DRAWN	TKS	08/05/2020
CHECKED	ABD	10/05/2020
ENGINEER	ABD	10/05/2020
APPROVED	SBA/JPD	12/05/2020
EMPLOYER		
SCALE	1:75	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
CRUSHING
DUCT-DUST EXTRACTION SYSTEM FOR TERTIARY CRUSHER HOUSE

DRAWING NUMBER: MD-216-3100-EG-ME-GAD-1028 | REV: A02